HACKATHON 2026

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TEAM LUMINA

Round: II

IDEA SUBMISSION

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Team Details:

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Track - Problem Statement:

Track Chosen: Coordinated AI Interview Panel (PS11)

- **Problem Statement (PS11):**

Build an adaptive voice interview platform in which one or more AI interviewers represent different roles. The interview panel may include a technical interviewer, hiring manager, customer, product manager or behavioural interviewer. The system should adapt its questions based on the candidate's previous answers rather than following only a predefined question list. The solution should demonstrate Real-time and interruptible voice interviews Multiple interviewer roles or personalities Shared candidate context between interviewer roles Dynamic follow-up questions Controlled interviewer turn-taking Role-play or scenario-based questions Difficulty adjustment based on candidate performance Identification of vague or contradictory answers Evidence-based feedback linked to the interview transcript A structured final assessment Clear disclosure that the candidate is interacting with AI Example scenario A candidate provides a technically correct solution but does not explain its impact on customers. The technical interviewer may accept the implementation, while the product interviewer should challenge the candidate to explain the business implications.

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Problem Description:

Most online interviews are run by single interviewer, so candidates are assessed from one narrow perspective. A real panel, technical, product and behavioural together, gives much stronger signal, but coordinating several senior interviewers for every candidate is slow, expensive, and hard to schedule, so early screening stays shallow and inconsistent.

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Proposed Solution / Idea Overview:

We are building a conducted multi-persona AI interview panel that runs entirely over live voice. Instead of single chatbot, three AI interviewers, a technical, a product and a behavioural persona, each with distinct voice, interview the candidate together in real time. A conductor decides who speaks, so exactly one interviewer holds the floor and can hand off to another with a reason: when candidate gives technically correct answer but never mentions customer impact, the technical interviewer accepts it and the product interviewer challenges the business gap. The panel shares context across roles, adapts difficulty to performance, catches vague or contradictory answers, supports interruption, and produces a structured, transcript-cited assessment. The panel size is configurable: we build with three roles and can add others such as a hiring manager without changing the architecture. Built on Agora's Conversational AI Engine, it runs on free tiers, delivering panel-quality screening that scales.

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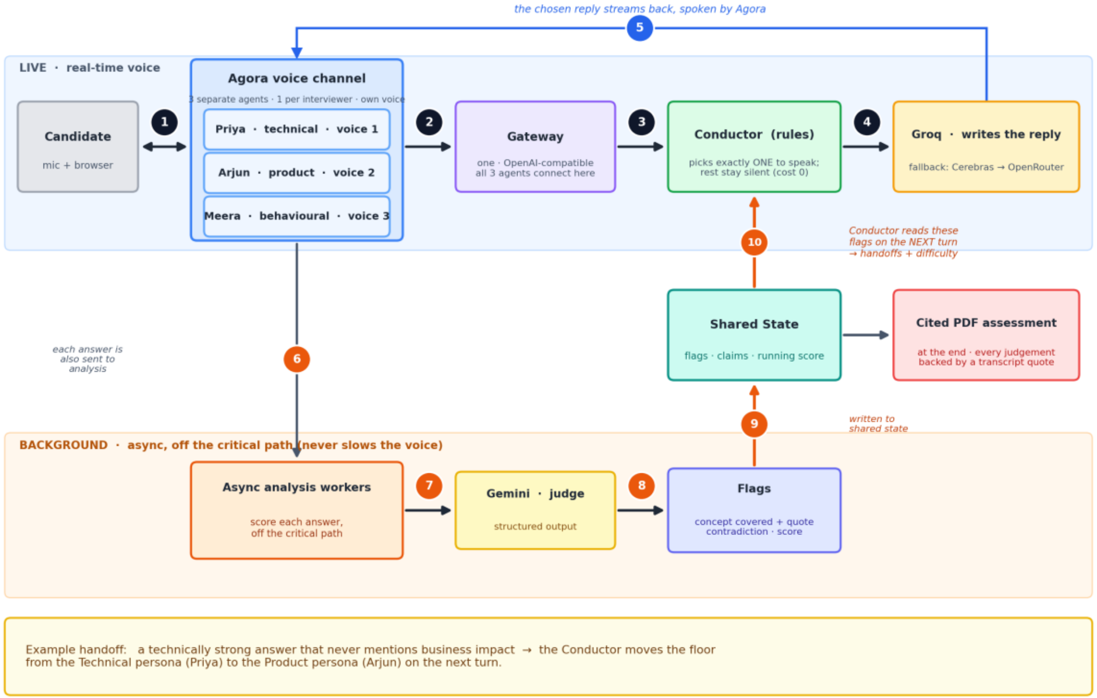
One interview turn: the LIVE row is what the candidate hears; the BACKGROUND row scores each answer and steers the next turn.



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Technical Architecture

- Three AI interviewers run as separate Agora voice agents, each with its own voice, behind one OpenAI-compatible gateway.
- Each turn, a rule-based conductor picks exactly one persona to speak; the rest stay silent, so only one voice is heard and silent personas cost nothing.
- The chosen persona's reply is generated by Groq (with Cerebras / OpenRouter fallback) and streamed back through Agora as speech.
- Off the critical path, async workers use Gemini to judge each answer into flags (concept covered plus quote, contradiction, score) and write them to shared state.
- The conductor reads those flags on the next turn to drive handoffs and difficulty, for example a technically strong answer that ignores business impact hands the floor to the product persona.
- At the end, state becomes a cited PDF assessment, every judgement backed by a transcript quote.

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Scoring & Assessment

Every candidate is scored out of a fixed total, allocated top-down so the marking is transparent and tunable per role:

- **Category weighting :** the total is split across interview categories (technical, product, behavioural) by how important each is for the role. Equal weighting to start, adjustable per job.
- **Total & normalization :** marks accumulate in the running score and are totalled just before the feedback step. Each answer is scored as a percentage of the marks available to it, and the total is normalized to a fixed standard, so results stay fair and comparable across candidates and interview lengths (an interview with fewer questions isn't under-counted).
- **Per question :** each interviewer's share is divided evenly across the questions they ask. Example: 25 marks over 5 questions = 5 marks each.
- **Main vs follow-up :** if a question has a follow-up, its marks split between the two, e.g. 3 for the main answer and 2 for the follow-up.
- **Difficulty weighting :** each question's marks are scaled by its difficulty band (easy, medium, hard), so reaching and clearing harder questions scores higher. Reuses the difficulty level the conductor already tracks.
- **Rubric judging :** each answer is scored by an LLM judge against a per-question rubric, marking which concepts were covered and quoting the proof from the transcript.
- **Penalties for vague or contradictory answers :** a capped deduction is applied when the analysis workers flag a dodge or a contradiction, backed by the flagged quote. Reuses the vagueness and contradiction flags already produced.

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Expected Impact

- **Panel interviews are the real bottleneck :** coordinating a [1] 3-6 person panel takes 4-7 hours of scheduling per candidate, and senior calendars rarely align.
- **Most candidates never get properly assessed :** companies lose up to [2] 89% of candidates to slow screening, and only [3] ~3% of ~180 applicants per hire reach an interview.
- **Hiring is slow and expensive :** [4]~$4,700-$5,500 per hire, [3] ~44 days to fill, with recruiters spending two-thirds of that time on interviews.
- **Multilinguality** widens access across the Indian talent market.
- **Our impact :** an always-available AI panel gives every candidate an instant, consistent, panel-quality interview with zero scheduling, reclaiming scarce senior time and cutting drop-off.

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Sources: [4] SHRM Benchmarking (2024-25); [1] US Tech Automations ; [2] G2; [3] high5test;

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Planned Utilization of Agora Technologies

- Agora Conversational AI Engine powers real-time speech-to-text, text-to-speech, turn detection, and barge-in.
- Multi-agent join: one Agora agent per interviewer persona, each with a distinct Agora voice.
- Each agent's llm.url points to our OpenAI-compatible gateway (llm.style = openai).
- Real-time interruption: candidates can talk over any interviewer mid-sentence.
- Agora endpoints used: join, interrupt, update, history, and per-turn latency metrics.
- Multilingual ASR and TTS voices for interviews in English , Indian (12+) and other languages

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Additional Notes:

- Differentiation: most entries are a single-interviewer bot; ours is a conducted panel with floor control, cross-persona memory, contradiction detection, and a cited report.
- Tech stack: Agora + Groq (Llama 3.3 70B) with Cerebras / OpenRouter fallback + Gemini + FastAPI, all on free tiers.
- Feasibility: designed to be built and demoed within the one-week sprint, with core scoring modules already prototyped.

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